# Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

***Ringkasan*—This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60**

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

graphicx
url
amsmath

***Ringkasan*—The short abstract (50-80 words) is intended to give the reader an overview of the work.**

## I. Introduction

This section introduces the topic and leads the reader on to the main part.

## II. Main Part

The presentation's content is summarized in the report in 4 pages. The author should fill, but not exceed, this space.
The report should be a self-contained report, so that it can be understood without studying additional literature.

## III. Format

The report can be written in LaTeX or Microsoft Word, but LaTeX is definitely preferred. Its appearance should be as close to this document as possible to achieve consistency in the proceedings.

References should be cited as numbers, and should be ordered by their appearance (example: "... as shown in [1], ..."). Only references that are

actually cited can be listed in the references section. The references' format should be evident from the examples in this text.

References should be of academic character and should be published and accessible. Your advisor can answer your questions regarding literature research. You must cite all used sources. Examples of good references include text books and scientific journals or conference proceedings. If possible, citing internet pages should be avoided. In particular, Wikipedia is *not* an appropriate reference in academic reports. Avoiding references in languages other than English is recommended.

Figures and tables should be labeled and numbered, such as in Table I and Fig. 1.

Tabel I
SIMULATION PARAMETERS

| | |
|---|---|
| Information message length | $k = 16000$ bit |
| Radio segment size | $b = 160$ bit |
| Rate of component codes | $R_{cc} = 1/3$ |
| Polynomial of component encoders | $[1, 33/37, 25/37]_8$ |

Gambar 1. Simulation results on the AWGN channel. Average throughput $k/n$ vs $E_s/N_0$.

## IV. FILLING THIS PAGE

Gallia est omnis divisa in partes tres, quarum unam incolunt Belgae, aliam Aquitani, tertiam qui ipsorum lingua Celtae, nostra Galli appellantur. Gallos ab Aquitanis Garumna flumen, a Belgis Matrona et Sequana dividit. Horum omnium fortissimi sunt Belgae, propterea quod a cultu atque humanitate provinciae longissime absunt, minimeque ad eos mercatores saepe commeant atque ea quae ad effeminandos animos pertinent important, proximique sunt Germanis, qui trans Rhenum incolunt, quibuscum continenter bellum gerunt. Qua de causa Helvetii quoque reliquos Gallos virtute praecedunt, quod fere cotidianis proeliis cum Germanis contendunt, cum aut suis finibus eos prohibent aut ipsi in eorum finibus bellum gerunt. Eorum una, pars, quam Gallos obtinere dictum est, initium capit a flumine Rhodano, continetur Garumna flumine, Oceano, finibus Belgarum, attingit etiam ab Sequanis et Helvetiis flumen Rhenum, vergit ad septentriones. Belgae ab extremis Galliae finibus oriuntur, pertinent ad inferiorem partem fluminis Rheni, spectant in septentrionem et orientem solem.

## V. CONCLUSION

This section summarizes the paper.

## PUSTAKA

[1] J. Hagenauer, E. Offer, and L. Papke. Iterative decoding of binary block and convolutional codes. *IEEE Trans. Inform. Theory*, vol. 42, no. 2, pp. 429â€“-445, Mar. 1996.

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[4] F. R. Kschischang. Giving a talk: Guidelines for the Preparation and Presentation of Technical Seminars. http://www.comm.toronto.edu/frank/guide/guide.pdf.

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